

Self-Attention vs. Masked Attention

Technical Summary Documentation

May 29, 2026

This document outlines the mechanics, mathematical foundations, and training paradigms of Self-Attention and Masked Self-Attention within Transformer architectures.

1 Architectural Context: Encoder vs. Decoder [0.8pt]

The core difference between standard self-attention and masked attention lies in the directionality of information flow, dictated by whether the model is designed to *understand* context or *generate* text.

- **Standard Self-Attention (Bi-directional):** Used in Transformer Encoders (e.g., BERT). Every token can attend to every other token in the sequence, regardless of whether they appear before or after. This is ideal for tasks requiring full contextual comprehension (e.g., classification, named entity recognition).
- **Masked Self-Attention (Causal / Uni-directional):** Used in Transformer Decoders (e.g., GPT, LLaMA). A token at position i is strictly forbidden from attending to any future tokens at positions $i + 1, i + 2, \dots, n$. This constraint is essential for autoregressive sequence generation.

2 Mathematical Foundations of Masked Attention [0.8pt]

In standard scaled dot-product attention, the operation is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \quad (1)$$

To enforce causality, a **Mask Matrix** (M) is injected into the formula immediately prior to the activation function:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}} + M\right) V \quad (2)$$

2.1 Step-by-Step Matrix Transformation

Consider a 4-token sequence during training: [BOS], The, cat, sat.

Step 1: Compute Raw Attention Scores $\left(\frac{QK^T}{\sqrt{d_k}}\right)$

The dot product of Queries (Q) and Keys (K) generates a 4×4 matrix representing unnormalized alignment scores between all token combinations.

$$\text{Raw Scores} = \begin{pmatrix} s_{11} & s_{12} & s_{13} & s_{14} \\ s_{21} & s_{22} & s_{23} & s_{24} \\ s_{31} & s_{32} & s_{33} & s_{34} \\ s_{41} & s_{42} & s_{43} & s_{44} \end{pmatrix} \quad (3)$$

Step 2: Apply the Causal Mask (M)

The mask matrix M is constructed with 0 on the main diagonal and lower triangle (allowed positions), and $-\infty$ (or a large negative number like $-1e9$) on the upper triangle (forbidden future positions).

$$M = \begin{pmatrix} 0 & -\infty & -\infty & -\infty \\ 0 & 0 & -\infty & -\infty \\ 0 & 0 & 0 & -\infty \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (4)$$

Adding M to the raw scores suppresses future positions:

$$\text{Masked Scores} = \begin{pmatrix} s_{11} & -\infty & -\infty & -\infty \\ s_{21} & s_{22} & -\infty & -\infty \\ s_{31} & s_{32} & s_{33} & -\infty \\ s_{41} & s_{42} & s_{43} & s_{44} \end{pmatrix} \quad (5)$$

Step 3: Softmax Normalization

The softmax function is evaluated row-wise. Because $e^{-\infty} \rightarrow 0$, the attention weights for any future tokens are driven exactly to zero.

$$\text{Attention Weights} = \begin{pmatrix} 1.0 & 0.0 & 0.0 & 0.0 \\ 0.4 & 0.6 & 0.0 & 0.0 \\ 0.2 & 0.3 & 0.5 & 0.0 \\ 0.1 & 0.2 & 0.3 & 0.4 \end{pmatrix} \quad (6)$$

- **Row 1 ([BOS]):** can only allocate attention to itself.
- **Row 3 (cat):** distributes its attention across [BOS], The, and cat, but is completely blind to sat.

3 The Learning Mechanism: Teacher Forcing & Parallel Training [0.8pt]

The inclusion of the attention mask allows the network to learn efficiently through a technique known as **Teacher Forcing** executed in parallel.

3.1 Parallel Forward Pass

During training, the ground-truth sequence is available all at once. Instead of looping through the sequence token-by-token, the entire sequence is passed through the network simultaneously. The mask creates distinct "blind spots" for each row:

Input Row (Model Processes)	Visible Context (Due to Mask)	Target Label (Truth)
Row 1: [BOS]	[BOS]	The
Row 2: [BOS] The	[BOS], The	cat
Row 3: [BOS] The cat	[BOS], The, cat	sat

Table 1: Visual context isolation during parallel forward pass.

3.2 Loss Computation

The model generates vocabulary-wide probability distributions for each row simultaneously. The Cross-Entropy Loss is calculated for every position relative to its respective target label:

$$\mathcal{L}_{\text{total}} = \frac{1}{N} \sum_{i=1}^N \mathcal{L}_i(y_i, \hat{y}_i) \quad (7)$$

Where y_i is the true target token and $\hat{y}_i$ is the model’s predicted distribution at step i .

3.3 Backpropagation Boundaries

During the backward pass, gradients flow from the total loss back through the attention weights to update W_Q , W_K , and W_V .

Because the attention weights for future tokens are zero, **no gradient information or error signals can flow from future positions back to past token representations**. This enforces a strict mathematical boundary: the weight matrices are updated solely based on how effectively they leverage *historical context* to predict the immediate *future token*.

4 Summary: Comparative Analysis

[0.8pt]

Feature	Standard Self-Attention	Masked Self-Attention
Directionality	Bi-directional (Past & Future)	Causal / Uni-directional (Past only)
Mask Matrix (M)	None (or padding masks only)	Upper-triangular values set to $-\infty$
Primary Use Case	Text Understanding / Encoding (BERT)	Text Generation / Decoding (GPT, LLaMA)
Training Efficiency	Fully parallelized	Fully parallelized via Teacher Forcing
Inference Behavior	Processes static context at once	Generates sequentially, step-by-step

Table 2: Key differences between Standard Self-Attention and Masked Attention.